

TRANSFORMATION BETWEEN CARTESIAN, CYLINDRICAL AND SPERICAL COORDINATION SYSTEM

Aim:

To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:

Test case (i) Transformation of Cartesian coordinate points in to Cylindrical and spherical

coordinate system

```
clc;
```

```
x=input('Enter the value of X=');
```

```
y=input('Enter the value of y=');
```

```
z=input('Enter the value of z=');
```

```
r=sqrt(x^2+y^2);
```

```
pi=atand(y/x);
```

```
disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=',[r pi z]);
```

```
r=sqrt(x^2+y^2+z^2);
```

```
theta=acosd(z/r);
```

```
pi=atand(y/x);
```

disp('Cartesian to spherical coordinate system of s(r,theta,pi)=[r theta pi]);

Output:-

Enter the value of X=13

Enter the value of y= -2

Enter the value of z=8

Practical output:

```
--> exec('C:\Users\jasiy\OneDrive\Documents\EXP 1 EME.sci', -1)
Enter the value of X=13
Enter the value of y=-2
Enter the value of z=8

"Cartesian to cylindrical coordinate system of p(r,pi,z)="
13.152946 -8.7461623 8.
"Cartesian to spherical coordinate system of s(r,theta,pi)="
15.394804 58.690831 -8.7461623
```

Theoretical output:

Theoretical :

Case (r) :

Rectangular $\rightarrow$ Cylindrical

$$\rho = \sqrt{x^2 + y^2} = \sqrt{(13)^2 + (-2)^2} = \sqrt{169 + 4}$$
$$= \sqrt{173} = 13.152$$

$\rho = 13.152$

$$\phi = \tan^{-1}(y/x)$$
$$= \tan^{-1}(-2/13) = \tan^{-1}(-0.153)$$

$\phi = -8.6987$

$$z = z$$

$z = 8$

Rectangular $\rightarrow$ Spherical

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{(13)^2 + (-2)^2 + (8)^2} = \sqrt{169 + 4 + 64}$$
$$r = \sqrt{237}$$
$$r = 15.39480432$$

$$\theta = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$

$$= \cos^{-1} \left(\frac{+8}{15.3948} \right)$$

$$\theta = 58.69082$$

$$\phi = \tan^{-1} (y/x) = \tan^{-1} (-2/18)$$

$$\phi = -8.746162263$$

Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system

clc;

r=input('Enter the value of r=');

pi=input('Enter the value of pi=');

z=input('Enter the value of z=');

x=r*cosd(pi);

y=r*sind(pi);

z=z;

disp('cylindrical to cartesian coordinate system of p(x,y,z)=',[x y z]);

r=sqrt(r^2+z^2);

pi=pi;

theta=acosd(z/r);

disp('cylindrical to spherical coordinate system of s(r,pi,theta)=',[r pi theta]);

Output:-

Enter the value of $r=5$

Enter the value of $\pi=75$

Enter the value of $z=-3$

Practical output:

```
--> exec('C:\Users\jasiy\OneDrive\Documents\EXp 1)ii.sci', -1)
Enter the value of r=5

Enter the value of pi=75

Enter the value of z=-3

"cylindrical to cartesian coordinate system of p(x,y,z)="
1.2940952  4.8296291  -3.
"cylindrical to spherical coordinate system of s(r,pi,theta)="
5.8309519  75.  120.96376
```

Theoretical output:

$\phi = -8.746162263$

Case (i) :
Output :
Enter the value of $r = 5$
Enter the value of $\pi = 75^\circ$
Enter the value of $z = -3$

"Cylindrical to cartesian coordinate system of $p(x,y,z) =$
1.2940952 4.826291 -3.

"Cylindrical to spherical coordinate system of $s(r,\pi,\theta) =$
5.8309519 75. 120.96376

Theoretical :
Case (i) :
Cylindrical $\rightarrow$ Rectangular
 $x = r \cos \phi$
 $x = 5 \cos 75^\circ$
 $x = 1.294095226$
 $y = r \sin \phi$
 $y = 5 \sin 75^\circ$
 $y = 4.829629131$
 $z = -3$

Cylindrical to Spherical

$$r = \sqrt{e^2 + z^2} = \sqrt{(5)^2 + (-3)^2} = \sqrt{25+9} = \sqrt{34}$$

$$r = 5.830951895$$

$$\phi = \tan^{-1}(e/z) = \tan^{-1}(5/-3)$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{-3}{5.830}\right)$$

$$\theta = 120.9693697$$

$$\phi = \phi$$

$$\phi = 75^\circ$$

Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system

clc;

r=input('Enter the value of r=');

pi=input('Enter the value of pi=');

theta=input('Enter the value of theta=');

x=r*sind(theta)*cos(pi);

y=r*sind(theta)*sin(pi);

z=r*cosd(theta);

disp('spherical to cartesian coordinate system of p(x,y,z)=[x y z];

p=r*sin(theta);

pi=pi;

z=r*cos(pi);

disp('Spherical to cylindrical coordinate system of s(p,pi,z)=[p pi z];')

Practical output:

```
Scilab 2026.0.1 Console
File Edit Control Applications ?
Scilab 2026.0.1 Console

Startup execution:
loading initial environment

--> exec('C:\Users\jasiy\OneDrive\Documents\exp 1\iii_eme.sci', -1)
Enter the value of r=7

Enter the value of pi=120

Enter the value of theta=100

"spherical to cartesian coordinate system of p(x,y,z)="
5.6126821 4.0025328 -1.2155372
"spherical to cylindrical coordinate system of s(p,pi,z)="
-3.5445595 120. 5.6992668

-->
```

Theoretical output:

Case (iii) :

Output :

Enter the value of $r = 7$

Enter the value of $\phi = 120^\circ$

Enter the value of $\theta = 100^\circ$

"Spherical to cartesian coordinate system of $p(x,y,z) =$ "

5.6126821 4.0025328 -1.2155372

"Spherical to cylindrical coordinate system of $s(p,\pi,z) =$ "

-3.5445595 120. 5.6992668

Theoretical :

Case (iii) :

Spherical $\rightarrow$ Rectangular

$$x = r \cos \phi = 7 \cos(120^\circ) = r \sin \theta \cos \phi = 7 \sin(100^\circ) \cos(120^\circ)$$
$$x = 5.61268$$
$$y = r \sin \phi \sin \theta = 7 \sin(100^\circ) \sin(120^\circ)$$
$$y = 4.002$$
$$z = r \cos \theta = 7 \cos(100^\circ) = -1.21553724$$

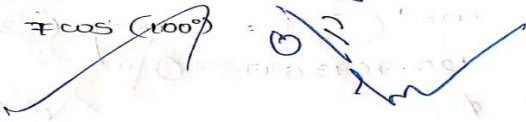
Spherical $\rightarrow$ Cylindrical

$$\rho = r \sin \theta = 7 \sin(100^\circ)$$

$$\rho = -3.54456$$

$$\phi = \phi = 120^\circ$$

$$z = r \cos \theta = 7 \cos(100^\circ)$$

$$z = 5.569926$$


Result:

Thus, the program was verified Transformation between Cartesian and Cylindrical coordinate system its

classified two categories (a) Cartesian to cylindrical and spherical coordinate system (b) Cylindrical to

cartesian and spherical coordinate system (c) spherical to cartesian and cylindrical coordinate system

using SCILAB was successfully.

